Extra Exercise 4

Find parametric equations for the tangent line to the curve with the given parametric equations at the specified point.

(a) $x = t^2 + 1, \quad y = 4\sqrt{t}, \quad z = e^{t^2 - t}$; $(2, 4, 1)$

(b) $x = e^t \cos t, \quad y = e^t \sin t, \quad z = e^{-t}$; $(1, 0, 1)$.